WEYLAND-YUTANI CORPORATION

Mining output analysis

2099-11-02 to 2100-01-30 | 90 days | 3 mines

Data source

Google Sheets | Snapshot ed961442f8fc | Loaded 2026-10-27T13:06:29+00:00

https://docs.google.com/spreadsheets/d/1u0EnnQ3ig7PLiot_ufticVRWXXeWjn1_cL1YOjQjPc4/edit

Daily statistics (tonnes)

Series	Mean	Std. dev.	Median	IQR	Flagged
LV-426	47.90	15.81	51.42	18.89	22
Origae-6	42.29	16.94	44.00	19.42	29
Fiorina 151	51.40	19.52	52.38	25.49	32
Total	141.59	45.77	155.96	42.56	25

Standard deviation uses the sample formula (n - 1). Quartiles use linear interpolation.

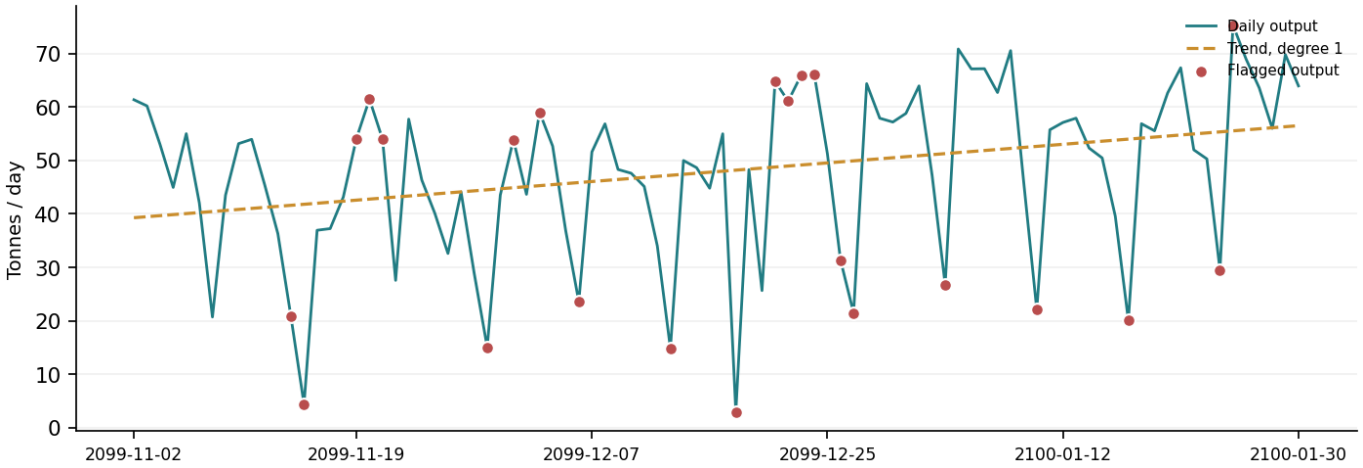
Analysis settings

IQR multiplier: 1.5; z-score threshold: 3.0; moving average: previous 7 days, 40.0% threshold; Grubbs alpha: 0.05; polynomial degree: 1; chart: line; highlighted method: any.

Grubbs is a two-sided single-outlier test assuming approximately normal, independent observations. Trends, weekly patterns and correlation can violate that assumption; its flags are exploratory for this generated series. No observations are removed.

Methods: IQR fences are Q1 - k*IQR and Q3 + k*IQR. Z-score is measured from the series mean. The moving-average baseline excludes the current day and starts after a full window. A positive value after a zero baseline is flagged.

LV-426



Detection counts

IQR: 2, Z-score: 0, Moving average: 22, Grubbs' test: 0

Total output: 4,311.35 tonnes. Mean daily output: 47.90 tonnes.

Grubbs G = 2.8444; critical value = 3.3477.

Anomaly details (22)

1. 2099-11-14 - Drop

Observed: 20.78 t. Baseline (previous 7 days): 42.16 t. Difference: -50.7%.

Flagged by: Moving average. Z-score: -1.715.

2. 2099-11-15 - Drop

Observed: 4.34 t. Baseline (previous 7 days): 39.12 t. Difference: -88.9%.

Flagged by: IQR, Moving average. Z-score: -2.755.

3. 2099-11-19 - Spike

Observed: 54.09 t. Baseline (previous 7 days): 32.02 t. Difference: +68.9%.

Flagged by: Moving average. Z-score: 0.391.

4. 2099-11-20 - Spike

Observed: 61.49 t. Baseline (previous 7 days): 33.27 t. Difference: +84.8%.

Flagged by: Moving average. Z-score: 0.859.

5. 2099-11-21 - Spike

Observed: 53.98 t. Baseline (previous 7 days): 36.86 t. Difference: +46.4%.

Flagged by: Moving average. Z-score: 0.384.

6. 2099-11-29 - Drop

Observed: 15.03 t. Baseline (previous 7 days): 39.63 t. Difference: -62.1%.

Flagged by: Moving average. Z-score: -2.079.

7. 2099-12-01 - Spike

Observed: 53.90 t. Baseline (previous 7 days): 35.81 t. Difference: +50.5%.
Flagged by: Moving average. Z-score: 0.379.

8. 2099-12-03 - Spike

Observed: 59.06 t. Baseline (previous 7 days): 37.40 t. Difference: +57.9%.
Flagged by: Moving average. Z-score: 0.705.

9. 2099-12-06 - Drop

Observed: 23.57 t. Baseline (previous 7 days): 43.53 t. Difference: -45.9%.
Flagged by: Moving average. Z-score: -1.539.

10. 2099-12-13 - Drop

Observed: 14.80 t. Baseline (previous 7 days): 43.88 t. Difference: -66.3%.
Flagged by: Moving average. Z-score: -2.093.

11. 2099-12-18 - Drop

Observed: 2.92 t. Baseline (previous 7 days): 41.78 t. Difference: -93.0%.
Flagged by: IQR, Moving average. Z-score: -2.844.

12. 2099-12-21 - Spike

Observed: 64.83 t. Baseline (previous 7 days): 39.33 t. Difference: +64.8%.
Flagged by: Moving average. Z-score: 1.070.

13. 2099-12-22 - Spike

Observed: 61.11 t. Baseline (previous 7 days): 41.45 t. Difference: +47.4%.
Flagged by: Moving average. Z-score: 0.835.

14. 2099-12-23 - Spike

Observed: 66.02 t. Baseline (previous 7 days): 43.23 t. Difference: +52.7%.
Flagged by: Moving average. Z-score: 1.146.

15. 2099-12-24 - Spike

Observed: 66.08 t. Baseline (previous 7 days): 46.26 t. Difference: +42.8%.
Flagged by: Moving average. Z-score: 1.149.

16. 2099-12-26 - Drop

Observed: 31.37 t. Baseline (previous 7 days): 54.75 t. Difference: -42.7%.
Flagged by: Moving average. Z-score: -1.045.

17. 2099-12-27 - Drop

Observed: 21.46 t. Baseline (previous 7 days): 52.33 t. Difference: -59.0%.
Flagged by: Moving average. Z-score: -1.672.

18. 2100-01-03 - Drop

Observed: 26.76 t. Baseline (previous 7 days): 53.00 t. Difference: -49.5%.
Flagged by: Moving average. Z-score: -1.337.

19. 2100-01-10 - Drop

Observed: 22.08 t. Baseline (previous 7 days): 58.75 t. Difference: -62.4%.
Flagged by: Moving average. Z-score: -1.633.

20. 2100-01-17 - Drop

Observed: 20.06 t. Baseline (previous 7 days): 47.90 t. Difference: -58.1%.
Flagged by: Moving average. Z-score: -1.761.

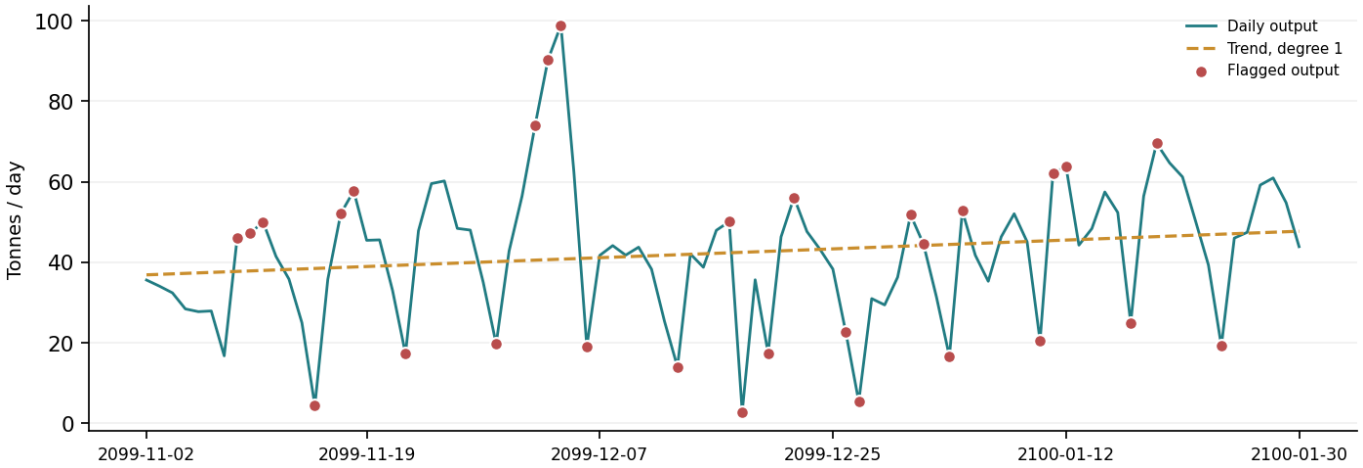
21. 2100-01-24 - Drop

Observed: 29.51 t. Baseline (previous 7 days): 52.12 t. Difference: -43.4%.
Flagged by: Moving average. Z-score: -1.163.

22. 2100-01-25 - Spike

Observed: 75.34 t. Baseline (previous 7 days): 53.47 t. Difference: +40.9%.
Flagged by: Moving average. Z-score: 1.735.

Origae-6



Detection counts

IQR: 3, Z-score: 1, Moving average: 29, Grubbs' test: 0

Total output: 3,806.14 tonnes. Mean daily output: 42.29 tonnes.

Grubbs G = 3.3389; critical value = 3.3477.

Anomaly details (29)

1. 2099-11-09 - Spike

Observed: 46.01 t. Baseline (previous 7 days): 28.99 t. Difference: +58.7%.

Flagged by: Moving average. Z-score: 0.220.

2. 2099-11-10 - Spike

Observed: 47.18 t. Baseline (previous 7 days): 30.48 t. Difference: +54.8%.

Flagged by: Moving average. Z-score: 0.289.

3. 2099-11-11 - Spike

Observed: 50.03 t. Baseline (previous 7 days): 32.35 t. Difference: +54.7%.

Flagged by: Moving average. Z-score: 0.457.

4. 2099-11-15 - Drop

Observed: 4.42 t. Baseline (previous 7 days): 37.46 t. Difference: -88.2%.

Flagged by: Moving average. Z-score: -2.235.

5. 2099-11-17 - Spike

Observed: 52.08 t. Baseline (previous 7 days): 34.24 t. Difference: +52.1%.

Flagged by: Moving average. Z-score: 0.578.

6. 2099-11-18 - Spike

Observed: 57.70 t. Baseline (previous 7 days): 34.94 t. Difference: +65.2%.

Flagged by: Moving average. Z-score: 0.910.

7. 2099-11-22 - Drop

Observed: 17.24 t. Baseline (previous 7 days): 39.14 t. Difference: -56.0%.
Flagged by: Moving average. Z-score: -1.479.

8. 2099-11-29 - Drop

Observed: 19.69 t. Baseline (previous 7 days): 45.17 t. Difference: -56.4%.
Flagged by: Moving average. Z-score: -1.334.

9. 2099-12-02 - Spike

Observed: 73.89 t. Baseline (previous 7 days): 44.37 t. Difference: +66.5%.
Flagged by: Moving average. Z-score: 1.865.

10. 2099-12-03 - Spike

Observed: 90.26 t. Baseline (previous 7 days): 46.32 t. Difference: +94.8%.
Flagged by: IQR, Moving average. Z-score: 2.831.

11. 2099-12-04 - Spike

Observed: 98.86 t. Baseline (previous 7 days): 52.30 t. Difference: +89.0%.
Flagged by: IQR, Z-score, Moving average. Z-score: 3.339.

12. 2099-12-06 - Drop

Observed: 19.04 t. Baseline (previous 7 days): 63.49 t. Difference: -70.0%.
Flagged by: Moving average. Z-score: -1.372.

13. 2099-12-13 - Drop

Observed: 14.02 t. Baseline (previous 7 days): 36.29 t. Difference: -61.4%.
Flagged by: Moving average. Z-score: -1.669.

14. 2099-12-17 - Spike

Observed: 50.09 t. Baseline (previous 7 days): 35.73 t. Difference: +40.2%.
Flagged by: Moving average. Z-score: 0.460.

15. 2099-12-18 - Drop

Observed: 2.84 t. Baseline (previous 7 days): 36.64 t. Difference: -92.2%.
Flagged by: IQR, Moving average. Z-score: -2.328.

16. 2099-12-20 - Drop

Observed: 17.43 t. Baseline (previous 7 days): 33.05 t. Difference: -47.3%.
Flagged by: Moving average. Z-score: -1.467.

17. 2099-12-22 - Spike

Observed: 56.10 t. Baseline (previous 7 days): 34.14 t. Difference: +64.3%.
Flagged by: Moving average. Z-score: 0.815.

18. 2099-12-26 - Drop

Observed: 22.63 t. Baseline (previous 7 days): 40.67 t. Difference: -44.4%.
Flagged by: Moving average. Z-score: -1.160.

19. 2099-12-27 - Drop

Observed: 5.40 t. Baseline (previous 7 days): 38.81 t. Difference: -86.1%.
Flagged by: Moving average. Z-score: -2.177.

20. 2099-12-31 - Spike

Observed: 51.80 t. Baseline (previous 7 days): 29.47 t. Difference: +75.8%.
Flagged by: Moving average. Z-score: 0.561.

21. 2100-01-01 - Spike

Observed: 44.49 t. Baseline (previous 7 days): 30.69 t. Difference: +45.0%.
Flagged by: Moving average. Z-score: 0.130.

22. 2100-01-03 - Drop

Observed: 16.59 t. Baseline (previous 7 days): 32.82 t. Difference: -49.5%.
Flagged by: Moving average. Z-score: -1.517.

23. 2100-01-04 - Spike

Observed: 52.73 t. Baseline (previous 7 days): 34.42 t. Difference: +53.2%.
Flagged by: Moving average. Z-score: 0.616.

24. 2100-01-10 - Drop

Observed: 20.46 t. Baseline (previous 7 days): 41.40 t. Difference: -50.6%.
Flagged by: Moving average. Z-score: -1.288.

25. 2100-01-11 - Spike

Observed: 62.15 t. Baseline (previous 7 days): 41.95 t. Difference: +48.2%.
Flagged by: Moving average. Z-score: 1.172.

26. 2100-01-12 - Spike

Observed: 63.82 t. Baseline (previous 7 days): 43.29 t. Difference: +47.4%.
Flagged by: Moving average. Z-score: 1.271.

27. 2100-01-17 - Drop

Observed: 24.88 t. Baseline (previous 7 days): 49.83 t. Difference: -50.1%.
Flagged by: Moving average. Z-score: -1.028.

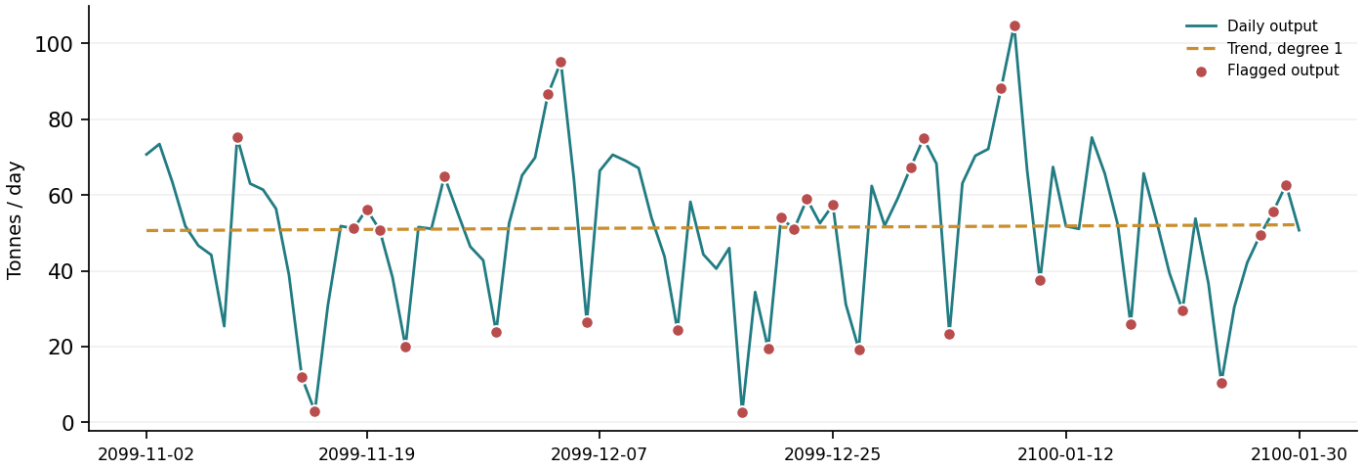
28. 2100-01-19 - Spike

Observed: 69.53 t. Baseline (previous 7 days): 49.64 t. Difference: +40.1%.
Flagged by: Moving average. Z-score: 1.608.

29. 2100-01-24 - Drop

Observed: 19.22 t. Baseline (previous 7 days): 52.33 t. Difference: -63.3%.
Flagged by: Moving average. Z-score: -1.362.

Fiorina 151



Detection counts

IQR: 1, Z-score: 0, Moving average: 32, Grubbs' test: 0

Total output: 4,625.82 tonnes. Mean daily output: 51.40 tonnes.

Grubbs G = 2.7384; critical value = 3.3477.

Anomaly details (32)

1. 2099-11-09 - Spike

Observed: 75.22 t. Baseline (previous 7 days): 53.67 t. Difference: +40.2%.

Flagged by: Moving average. Z-score: 1.220.

2. 2099-11-14 - Drop

Observed: 11.86 t. Baseline (previous 7 days): 52.09 t. Difference: -77.2%.

Flagged by: Moving average. Z-score: -2.026.

3. 2099-11-15 - Drop

Observed: 2.90 t. Baseline (previous 7 days): 47.47 t. Difference: -93.9%.

Flagged by: Moving average. Z-score: -2.485.

4. 2099-11-18 - Spike

Observed: 51.17 t. Baseline (previous 7 days): 36.30 t. Difference: +41.0%.

Flagged by: Moving average. Z-score: -0.012.

5. 2099-11-19 - Spike

Observed: 56.28 t. Baseline (previous 7 days): 34.84 t. Difference: +61.6%.

Flagged by: Moving average. Z-score: 0.250.

6. 2099-11-20 - Spike

Observed: 50.85 t. Baseline (previous 7 days): 34.82 t. Difference: +46.0%.

Flagged by: Moving average. Z-score: -0.028.

7. 2099-11-22 - Drop

Observed: 19.99 t. Baseline (previous 7 days): 40.29 t. Difference: -50.4%.
Flagged by: Moving average. Z-score: -1.609.

8. 2099-11-25 - Spike

Observed: 64.94 t. Baseline (previous 7 days): 45.60 t. Difference: +42.4%.
Flagged by: Moving average. Z-score: 0.694.

9. 2099-11-29 - Drop

Observed: 23.86 t. Baseline (previous 7 days): 47.48 t. Difference: -49.7%.
Flagged by: Moving average. Z-score: -1.411.

10. 2099-12-03 - Spike

Observed: 86.61 t. Baseline (previous 7 days): 50.89 t. Difference: +70.2%.
Flagged by: Moving average. Z-score: 1.804.

11. 2099-12-04 - Spike

Observed: 95.18 t. Baseline (previous 7 days): 55.32 t. Difference: +72.0%.
Flagged by: Moving average. Z-score: 2.243.

12. 2099-12-06 - Drop

Observed: 26.45 t. Baseline (previous 7 days): 65.37 t. Difference: -59.5%.
Flagged by: Moving average. Z-score: -1.278.

13. 2099-12-13 - Drop

Observed: 24.33 t. Baseline (previous 7 days): 56.77 t. Difference: -57.1%.
Flagged by: Moving average. Z-score: -1.387.

14. 2099-12-18 - Drop

Observed: 2.75 t. Baseline (previous 7 days): 44.46 t. Difference: -93.8%.
Flagged by: Moving average. Z-score: -2.492.

15. 2099-12-20 - Drop

Observed: 19.43 t. Baseline (previous 7 days): 35.80 t. Difference: -45.7%.
Flagged by: Moving average. Z-score: -1.638.

16. 2099-12-21 - Spike

Observed: 54.09 t. Baseline (previous 7 days): 35.10 t. Difference: +54.1%.
Flagged by: Moving average. Z-score: 0.138.

17. 2099-12-22 - Spike

Observed: 50.94 t. Baseline (previous 7 days): 34.51 t. Difference: +47.6%.
Flagged by: Moving average. Z-score: -0.023.

18. 2099-12-23 - Spike

Observed: 58.97 t. Baseline (previous 7 days): 35.46 t. Difference: +66.3%.
Flagged by: Moving average. Z-score: 0.388.

19. 2099-12-25 - Spike

Observed: 57.41 t. Baseline (previous 7 days): 39.02 t. Difference: +47.1%.
Flagged by: Moving average. Z-score: 0.308.

20. 2099-12-27 - Drop

Observed: 19.32 t. Baseline (previous 7 days): 46.37 t. Difference: -58.3%.
Flagged by: Moving average. Z-score: -1.643.

21. 2099-12-31 - Spike

Observed: 67.22 t. Baseline (previous 7 days): 47.70 t. Difference: +40.9%.
Flagged by: Moving average. Z-score: 0.811.

22. 2100-01-01 - Spike

Observed: 74.95 t. Baseline (previous 7 days): 49.79 t. Difference: +50.5%.
Flagged by: Moving average. Z-score: 1.207.

23. 2100-01-03 - Drop

Observed: 23.44 t. Baseline (previous 7 days): 57.60 t. Difference: -59.3%.
Flagged by: Moving average. Z-score: -1.432.

24. 2100-01-07 - Spike

Observed: 88.29 t. Baseline (previous 7 days): 62.79 t. Difference: +40.6%.
Flagged by: Moving average. Z-score: 1.890.

25. 2100-01-08 - Spike

Observed: 104.85 t. Baseline (previous 7 days): 65.80 t. Difference: +59.3%.
Flagged by: IQR, Moving average. Z-score: 2.738.

26. 2100-01-10 - Drop

Observed: 37.45 t. Baseline (previous 7 days): 69.83 t. Difference: -46.4%.
Flagged by: Moving average. Z-score: -0.715.

27. 2100-01-17 - Drop

Observed: 25.85 t. Baseline (previous 7 days): 57.26 t. Difference: -54.9%.
Flagged by: Moving average. Z-score: -1.309.

28. 2100-01-21 - Drop

Observed: 29.66 t. Baseline (previous 7 days): 53.86 t. Difference: -44.9%.
Flagged by: Moving average. Z-score: -1.114.

29. 2100-01-24 - Drop

Observed: 10.51 t. Baseline (previous 7 days): 43.45 t. Difference: -75.8%.
Flagged by: Moving average. Z-score: -2.095.

30. 2100-01-27 - Spike

Observed: 49.50 t. Baseline (previous 7 days): 34.67 t. Difference: +42.8%.
Flagged by: Moving average. Z-score: -0.097.

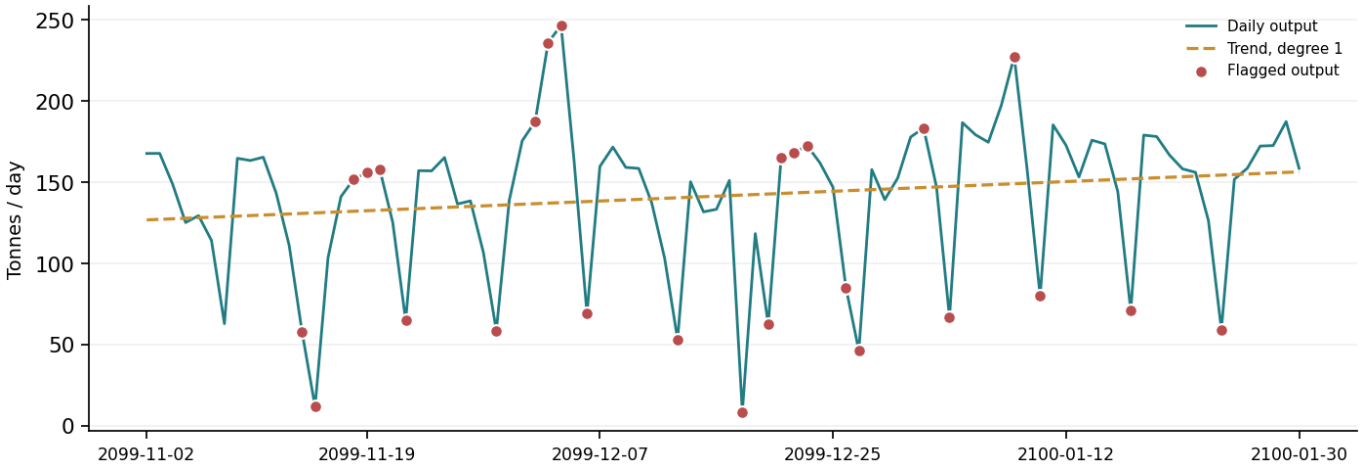
31. 2100-01-28 - Spike

Observed: 55.63 t. Baseline (previous 7 days): 36.12 t. Difference: +54.0%.
Flagged by: Moving average. Z-score: 0.217.

32. 2100-01-29 - Spike

Observed: 62.76 t. Baseline (previous 7 days): 39.83 t. Difference: +57.6%.
Flagged by: Moving average. Z-score: 0.582.

Total



Detection counts

IQR: 9, Z-score: 0, Moving average: 25, Grubbs' test: 0

Total output: 12,743.31 tonnes. Mean daily output: 141.59 tonnes.

Grubbs G = 2.9077; critical value = 3.3477.

Anomaly details (25)

1. 2099-11-14 - Drop

Observed: 57.62 t. Baseline (previous 7 days): 132.13 t. Difference: -56.4%.

Flagged by: IQR, Moving average. Z-score: -1.835.

2. 2099-11-15 - Drop

Observed: 11.66 t. Baseline (previous 7 days): 124.05 t. Difference: -90.6%.

Flagged by: IQR, Moving average. Z-score: -2.839.

3. 2099-11-18 - Spike

Observed: 152.02 t. Baseline (previous 7 days): 104.81 t. Difference: +45.0%.

Flagged by: Moving average. Z-score: 0.228.

4. 2099-11-19 - Spike

Observed: 155.82 t. Baseline (previous 7 days): 102.89 t. Difference: +51.4%.

Flagged by: Moving average. Z-score: 0.311.

5. 2099-11-20 - Spike

Observed: 157.90 t. Baseline (previous 7 days): 104.69 t. Difference: +50.8%.

Flagged by: Moving average. Z-score: 0.356.

6. 2099-11-22 - Drop

Observed: 64.79 t. Baseline (previous 7 days): 121.03 t. Difference: -46.5%.

Flagged by: Moving average. Z-score: -1.678.

7. 2099-11-29 - Drop

Observed: 58.58 t. Baseline (previous 7 days): 132.28 t. Difference: -55.7%.
Flagged by: IQR, Moving average. Z-score: -1.814.

8. 2099-12-02 - Spike

Observed: 187.43 t. Baseline (previous 7 days): 131.44 t. Difference: +42.6%.
Flagged by: Moving average. Z-score: 1.002.

9. 2099-12-03 - Spike

Observed: 235.93 t. Baseline (previous 7 days): 134.62 t. Difference: +75.3%.
Flagged by: IQR, Moving average. Z-score: 2.061.

10. 2099-12-04 - Spike

Observed: 246.73 t. Baseline (previous 7 days): 148.81 t. Difference: +65.8%.
Flagged by: IQR, Moving average. Z-score: 2.297.

11. 2099-12-06 - Drop

Observed: 69.06 t. Baseline (previous 7 days): 172.39 t. Difference: -59.9%.
Flagged by: Moving average. Z-score: -1.585.

12. 2099-12-13 - Drop

Observed: 53.15 t. Baseline (previous 7 days): 136.94 t. Difference: -61.2%.
Flagged by: IQR, Moving average. Z-score: -1.932.

13. 2099-12-18 - Drop

Observed: 8.51 t. Baseline (previous 7 days): 122.87 t. Difference: -93.1%.
Flagged by: IQR, Moving average. Z-score: -2.908.

14. 2099-12-20 - Drop

Observed: 62.51 t. Baseline (previous 7 days): 106.64 t. Difference: -41.4%.
Flagged by: Moving average. Z-score: -1.728.

15. 2099-12-21 - Spike

Observed: 165.23 t. Baseline (previous 7 days): 107.97 t. Difference: +53.0%.
Flagged by: Moving average. Z-score: 0.516.

16. 2099-12-22 - Spike

Observed: 168.15 t. Baseline (previous 7 days): 110.11 t. Difference: +52.7%.
Flagged by: Moving average. Z-score: 0.580.

17. 2099-12-23 - Spike

Observed: 172.61 t. Baseline (previous 7 days): 115.31 t. Difference: +49.7%.
Flagged by: Moving average. Z-score: 0.678.

18. 2099-12-26 - Drop

Observed: 85.19 t. Baseline (previous 7 days): 142.24 t. Difference: -40.1%.
Flagged by: Moving average. Z-score: -1.232.

19. 2099-12-27 - Drop

Observed: 46.18 t. Baseline (previous 7 days): 137.51 t. Difference: -66.4%.
Flagged by: IQR, Moving average. Z-score: -2.085.

20. 2100-01-01 - Spike

Observed: 183.41 t. Baseline (previous 7 days): 129.40 t. Difference: +41.7%.
Flagged by: Moving average. Z-score: 0.914.

21. 2100-01-03 - Drop

Observed: 66.79 t. Baseline (previous 7 days): 143.42 t. Difference: -53.4%.
Flagged by: Moving average. Z-score: -1.634.

22. 2100-01-08 - Spike

Observed: 227.46 t. Baseline (previous 7 days): 162.15 t. Difference: +40.3%.
Flagged by: Moving average. Z-score: 1.876.

23. 2100-01-10 - Drop

Observed: 79.99 t. Baseline (previous 7 days): 169.98 t. Difference: -52.9%.
Flagged by: Moving average. Z-score: -1.346.

24. 2100-01-17 - Drop

Observed: 70.79 t. Baseline (previous 7 days): 154.99 t. Difference: -54.3%.
Flagged by: Moving average. Z-score: -1.547.

25. 2100-01-24 - Drop

Observed: 59.24 t. Baseline (previous 7 days): 147.89 t. Difference: -59.9%.
Flagged by: IQR, Moving average. Z-score: -1.799.